Evaluation

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Metric selection

Now we need to select a metric for choosing the hyperparameters and evaluating the models after training. The first things that come to mind are overall accuracy and F1-score. But there is a problem. From out analysis of data from previous task, we know that the dataset is not balanced. It means, that regular metrics won't show the true results.

Metric selection

We decided to use balanced accuracy and Macro-F1 score. In those metrics scores of all classes will contribute equally.

Both scores have ranges from zero to one, where one is the best possible result.

$$\text{Balanced Accuracy} = \frac{1}{N}\sum_{i=0}^{N-1}\text{Recall}_i$$

$$\text{Macro-F1} = \frac{1}{N}\sum_{i=1}^{N}\text{F1}_i$$

Baseline classifier

Why? - To measure and compare the performance of trained models.

A Challenge. Just firing random classes uniformly won't help us much, as the dataset is not balanced.

How? - Our idea is to create a random classifier that takes the frequency of all classes and generates labels with the same frequency.

The performance of such classifier is dramatically low: the balanced accuracy is 0.5 and Macro F1 score 0.059. Perfect performance is not achievable for this kind of classifier, as random predictions will guess the real result only in rare cases.

Thank you for the attention!